Graphic Era

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: M.Tech(IIIrd Sem)
Name of the Paper: Information Retrieval and Natural Language Processing
Paper Code: MCS329

Time: 3 hour | Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) (CO1)

a. Discuss how the following properties of natural languages make NLP tasks challenging:

- Synonymy
- Polysemy
- Context-dependence

b. Explain the purpose of tokenization in NLP. Why is it a critical step in text preprocessing?

c. You are tasked with building a Named Entity Recognition (NER) model for identifying custom entities such as product names, prices, and discounts in e-commerce data. Discuss how you would collect labeled data, choose a model, and evaluate its performance.

Q2. (2X10=20 Marks) (CO2)

a. Compare and contrast syntactic and semantic analysis in NLP. Provide examples of tasks where each is critical.

b. Explain why the Bag of Words (BoW) model is considered a simplistic representation of text data. What are its limitations, and how can these be addressed?

c. Evaluate the trade-offs between stemming and lemmatization in real-world applications like search engines or document summarization. Which technique would you recommend and why?

Q3. (2X10=20 Marks) (CO3)

a. How can you preprocess textual data to convert it into features and labels for a classification task? Discuss one example using Bag-of-Words and another using word embeddings.

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c. Explain how to incorporate context-sensitive embeddings like BERT in a pipeline for converting text to features and labels. Discuss the trade-offs between computational efficiency and accuracy.

Q4. (2X10=20 Marks) (CO4)

a. Explain the role of an inverted index in information retrieval systems. How does it improve search efficiency compared to sequential search?

b. Describe the term frequency-inverse document frequency (TF-IDF) model and its importance in term weighting. How does TF-IDF help in ranking documents in the vector space model?

c. Explain the role of user intent in modern information retrieval systems. How would you integrate user behavior data, such as click-through rates and dwell time, into the ranking algorithm?

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Q5. (2X10=20 Marks) (CO5)

a. Discuss the importance of Normalized Discounted Cumulative Gain (nDCG) in evaluating ranked retrieval systems. How does it handle the position of relevant documents in a ranked list?

b. Explain how search engines combine multiple ranking signals (e.g., textual relevance, link-based metrics, and user behavior) to optimize document ranking. Discuss how machine learning models are used in this process.

c. Critically analyze the limitations of feedback techniques in adversarial environments, such as when spam documents are present. Propose solutions to mitigate these risks.